| SLAB | DIMENSIONS | TYPE OF SLAB | TYPE OF PANEL | AREA OF STEEL PROVIDED AT MID SPAN | | EXTRA BARS AT SUPPORTS |
|---|---|---|---|---|---|---|
| S2 | 3100X3300 | Two-way slab | Two adjacent edges discontinuous | SHORTER SPAN | 8mm dia @ 240mm c/c | 8mm dia @ 240mm c/c |
| | | | | LONGER SPAN | 8mm dia @ 210mm c/c | 8mm dia @ 210mm c/c |
| S3 | 2800X3280 | Two-way slab | One shorter edge discontinues | SHORTER SPAN | 8mm dia @ 240mm c/c | 8mm dia @ 240mm c/c |
| | | | | LONGER SPAN | 8mm dia @ 210mm c/c | 8mm dia @ 210mm c/c |
| S4 | 2100X3280 | Two-way slab | Two adjacent edges discontinuous | SHORTER SPAN | 8mm dia @ 240mm c/c | 8mm dia @ 240mm c/c |
| | | | | LONGER SPAN | 8mm dia @ 210mm c/c | 8mm dia @ 210mm c/c |
| S5 | 1920X3280 | Two-way slab | Three edges discontinuous | SHORTER SPAN | 8mm dia @ 240mm c/c | 8mm dia @ 240mm c/c |
| | | | | LONGER SPAN | 8mm dia @ 210mm c/c | 8mm dia @ 210mm c/c |

***Table 4.3:*** Roof slab details.

| Frame | Span | Beam Size (b × D) mm | Eff. Depth(d)mm | $M_u$ KN-m | $Ast_{req}$ $mm^2$ | Min Ast $mm^2$ | Ast Provided $mm^2$ | Main Steel Provided |
|---|---|---|---|---|---|---|---|---|
| A2-B2-C2 | A2-B2 | 230 × 300 mm | 370 | 24.75 | 161.43 | 144.67 | 226.20 | 2 bars of 12mm Ø |
| | B2-C2 | 230 × 300 mm | 370 | 32.30 | 214.16 | 144.67 | 226.20 | 2 bars of 12mm Ø |
| A3-B3-C3 | A3-B3 | 230 × 300 mm | 370 | 26.48 | 173.18 | 144.67 | 226.20 | 2 bars of 12mm Ø |
| | B3-C3 | 230 × 300 mm | 370 | 33.45 | 222.44 | 144.67 | 226.20 | 2 bars of 12mm Ø |
| A1-A2-A3 | A1-A2 | 230 × 300 mm | 370 | 39.15 | 263.67 | 144.67 | 339.29 | 3 bars of 12mm Ø |
| | A2-A3 | 230 × 300 mm | 370 | 48.86 | 336.90 | 144.67 | 339.29 | 3 bars of 12mm Ø |
| B1-B2-B3 | B1-B2 | 230 × 300 mm | 370 | 38.78 | 260.91 | 144.67 | 339.29 | 3 bars of 12mm Ø |
| | B2-B3 | 230 × 300 mm | 370 | 27.17 | 177.94 | 144.67 | 226.20 | 2 bars of 12mm Ø |
| C1-C2-C3 | C1-C2 | 230 × 300 mm | 370 | 39.54 | 266.53 | 144.67 | 339.29 | 3 bars of 12mm Ø |
| | C2-C3 | 230 × 300 mm | 370 | 37.98 | 255.22 | 144.67 | 339.29 | 3 bars of 12mm Ø |

***Table 4.4:*** Design of Main Reinforcement

| Frame | Span | Joint | Factored Shear (Vu) KN | Nom. Shear ($\tau_v$) | Conc. Strength ($\tau_c$) N/mm$^2$ | Shear For Steel ($V_{us}$) | Stirrup Dia & Legs | Spacing ($S_v$) mm | Max. Spacing | Final Provided Spacing |
|---|---|---|---|---|---|---|---|---|---|---|
| A2-B2-C2 | A2-B2 | A2 | 31.56 | 0.39 | 0.30 | 8.03 | 8 mm Ø & 2 Leg | 2014.36 | 277.5 | 275 |
| | | B2 | 37.17 | 0.44 | 0.37 | 5.68 | 8 mm Ø & 2 Leg | 2847.76 | 277.5 | 275 |
| | B2-C2 | B2 | 46.13 | 0.54 | 0.37 | 14.64 | 8 mm Ø & 2 Leg | 1104.87 | 277.5 | 275 |
| | | C2 | 37.20 | 0.44 | 0.30 | 11.67 | 8 mm Ø & 2 Leg | 1386.06 | 277.5 | 275 |
| A3-B3-C3 | A3-B3 | A3 | 38.54 | 0.45 | 0.30 | 13.01 | 8 mm Ø & 2 Leg | 1242.91 | 277.5 | 275 |
| | | B3 | 37.11 | 0.44 | 0.37 | 5.62 | 8 mm Ø & 2 Leg | 2877.27 | 277.5 | 275 |
| | B3-C3 | B3 | 48.02 | 0.56 | 0.37 | 16.53 | 8 mm Ø & 2 Leg | 978.24 | 277.5 | 275 |
| | | C3 | 38.30 | 0.45 | 0.30 | 12.77 | 8 mm Ø & 2 Leg | 1266.27 | 277.5 | 275 |
| A1-A2-A3 | A1-A2 | A1 | 40.98 | 0.48 | 0.30 | 15.45 | 8 mm Ø & 2 Leg | 1046.87 | 277.5 | 275 |
| | | A2 | 53.07 | 0.62 | 0.43 | 16.48 | 8 mm Ø & 2 Leg | 981.44 | 277.5 | 275 |
| | A2-A3 | A2 | 65.51 | 0.77 | 0.43 | 28.92 | 8 mm Ø & 2 Leg | 559.27 | 277.5 | 275 |
| | | A3 | 53.66 | 0.63 | 0.30 | 28.13 | 8 mm Ø & 2 Leg | 574.98 | 277.5 | 275 |
| B1-B2-B2 | B1-B2 | B1 | 42.21 | 0.50 | 0.30 | 16.68 | 8 mm Ø & 2 Leg | 969.45 | 277.5 | 275 |
| | | B2 | 50.94 | 0.60 | 0.37 | 19.45 | 8 mm Ø & 2 Leg | 831.39 | 277.5 | 275 |
| | B2-B3 | B2 | 37.77 | 0.44 | 0.37 | 6.28 | 8 mm Ø & 2 Leg | 2574.92 | 277.5 | 275 |
| | | B3 | 28.50 | 0.33 | 0.30 | 2.97 | 8 mm Ø & 2 Leg | 5444.62 | 277.5 | 275 |
| C1-C2-C3 | C1-C2 | C1 | 42.29 | 0.50 | 0.30 | 16.76 | 8 mm Ø & 2 Leg | 964.81 | 277.5 | 275 |
| | | C2 | 52.73 | 0.62 | 0.37 | 21.24 | 8 mm Ø & 2 Leg | 761.31 | 277.5 | 275 |
| | C2-C3 | C2 | 51.66 | 0.61 | 0.37 | 20.17 | 8 mm Ø & 2 Leg | 801.70 | 277.5 | 275 |
| | | C3 | 40.98 | 0.48 | 0.30 | 15.45 | 8 mm Ø & 2 Leg | 1046.62 | 277.5 | 275 |

***Table 4.5:*** Design of Shear Reinforcement (Stirrups)

| Column | Type | Size (mm) | $P_u$ (KN) | $M_{ux}$ (KN-m) | $M_{uy}$ (KN-m) | Required p (%) | $A_{sc}$ Req ($mm^2$) | Main Steel Provided | $A_{sc}$ Prov ($mm^2$) | Ties Provided |
|---|---|---|---|---|---|---|---|---|---|---|
| C1 | Bi-axial | 230 x 300 | 53.94 | 9.73 | 8.04 | 0.8 | 552 | 4 bars of 16 mm Ø | 804 | 8mm Ø @ 200mm c/c |
| A2 | Uni-axial | 230 x 300 | 109.43 | 2.48 | 2.19 | 0.8 | 552 | 4 bars of 16 mm Ø | 804 | 8mm Ø @ 230mm c/c |
| B2 | Bi-axial | 230 x 300 | 114.67 | 2.97 | 3.58 | 0.8 | 552 | 4 bars of 16 mm Ø | 804 | 8mm Ø @ 200mm c/c |
| C2 | Uni-axial | 230 x 300 | 83.94 | 1.68 | 7.31 | 0.8 | 552 | 4 bars of 16 mm Ø | 804 | 8mm Ø @ 230mm c/c |
| A3 | Bi-axial | 230 x 300 | 76.20 | 12.18 | 1.52 | 0.8 | 552 | 4 bars of 16 mm Ø | 804 | 8mm Ø @ 200mm c/c |
| B3 | Uni-axial | 230 x 300 | 75.75 | 6.32 | 4.88 | 0.8 | 552 | 4 bars of 16 mm Ø | 804 | 8mm Ø @ 230mm c/c |
| C3 | Bi-axial | 230 x 300 | 52.85 | 9.22 | 7.17 | 0.8 | 552 | 4 bars of 16 mm Ø | 804 | 8mm Ø @ 200mm c/c |

***Table 4.6:*** Design of Columns C1, A2, B2, C2, A3, B3 & C3

| Footing | Factored Load $P_u$ | Upward Pressure $q_u$ (KN./$m^2$) | Max Moment $M_u$ (KN-m) | One-Way Shear $V_{u1}$ (KN) | Punching Shear $V_{u2}$ (KN) | Status |
|---|---|---|---|---|---|---|
| B1 | 133.04 | 133.04 | 9.86 | 25.28 | 105.10 | SAFE |
| C1 | 53.94 | 53.94 | 4.00 | 10.25 | 42.61 | SAFE |
| A2 | 109.43 | 109.43 | 8.11 | 20.79 | 86.45 | SAFE |
| B2 | 114.67 | 114.67 | 8.50 | 21.79 | 90.59 | SAFE |
| C2 | 83.94 | 83.94 | 6.22 | 15.95 | 66.31 | SAFE |
| A3 | 76.20 | 76.20 | 5.65 | 14.48 | 60.20 | SAFE |
| B3 | 75.75 | 75.75 | 5.61 | 14.39 | 59.84 | SAFE |
| C3 | 52.85 | 52.85 | 3.92 | 10.04 | 41.75 | SAFE |

***Table 4.7:*** Design of footing B1, C1, A2, B2, C2, A3, B3 & C3